

CMI REFEREE FIREWALL REPORT

BSD Rank 0 Submission Isolation Verification | Opera Numerorum | Battle Plan v1.6

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CMI REFEREE FIREWALL REPORT

Subject: BSD Rank 0 Submission Isolation Verification

Date: 2026-06-06 06:59 UTC

Repositories Audited:

opera-numerorum [PUBLIC] ./ (this workspace)

morningstar_spacecraft [PRIVATE] ./MORNING_STAR_REPO/

CLAY Seal: 5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcbf7ebe3c9

1. IMPORT GRAPH: BSD has zero dependencies on spaceship

1a. Searching proofs/ for forbidden imports (MS/BDP/spacecraft/Z_Protocol)...

RESULT: PASS -- No forbidden imports found in proofs/

1b. Import list for proofs/ACT3/BSD_M22.lean:

(none -- no active import statements)

2. CONSTANT PROVENANCE: M^* = 4/55 derived, not copied

2a. M^* definition context in BSD_M22.lean:

6:

7: Lean 4 proof skeleton for the M^* normalisation theorem.

8:

9: SCOPE: BSD submission only.

10: IMPORTS: None from morningstar_spacecraft, MS Tower, BDP Tower, or Z Protocol.

...

2b. Confirming M_{star} is NOT hardcoded as 'def M_{star} := 4/55':

PASS -- M_{star} defined via off_cliff_ratio, not as literal 4/55.

Definition: M_{star} := off_cliff_ratio where off_cliff_ratio := 4/55

Provenance: off_cliff_ratio derived from m22.out (SHA 5a5a345f...)

2c. Theorem M22.1 (M_{star} = 4/55) located:

Line 83: theorem $M_{\text{star_eq_four_over_55}}$: M_{star} = (4 : $\blacksquare$) / 55 := by

Line 89: theorem $M_{\text{star_ne_naive}}$: M_{star} < (1 : $\blacksquare$) := by

3. SORRY QUARANTINE: proofs/ has SORRY:0; BDP sorries scoped and reported

3a. BSD submission scope -- proofs/ sorry count:

proofs/ sorry (active, non-comment): 0

RESULT: PASS -- BSD submission (proofs/) is SORRY: 0

3b. MORNING_STAR_REPO/ total sorry count:

Total active sorries in MORNING_STAR_REPO/: 8

Inside archive/HISTORICAL/ : 0

Outside archive/HISTORICAL/ : 8

FINDING: Sorries exist outside archive/HISTORICAL/:

FILE: MORNING_STAR_REPO/src/M_FINAL/BDP_PhaseReversal.lean

line 33: sorry

line 57: sorry

line 72: sorry

line 90: sorry

line 101: · sorry

line 131: · sorry

line 135: · sorry

line 146: sorry

SCOPE NOTE: All sorries above are in BDP_PhaseReversal.lean

(P vs NP Tower, src/M_FINAL/). These are annotated sorry-fills pointing to certified bdp1.out .. bdp4.out computations.

They are NOT in the BSD submission scope (proofs/).

BSD ISOLATION: PASS (zero sorries in proofs/)

MORNINGSTAR FULL QUARANTINE: PARTIAL (BDP tower sorries in src/M_FINAL/)

STRUCTURAL ISOLATION (June 2026):

MORNING_STAR_REPO/ now lives in a physically separate private GitHub repository (github.com/DavidFox998/morningstar_spacecraft). A Clay referee cloning opera-numerorum sees zero BDP/Morning Star files by construction. The sorry-isolation is structural (repo-level), not only import-graph-verified. Push scripts: push_to_github.sh (public) and push_morning_star.sh (private). See REPO_MAP.txt.

3c. STUB_MANIFEST contents:

STUB MANIFEST -- PVS_NP HISTORICAL ARCHIVE

Opera Numerorum / Battle Plan v1.6

Date archived: May 2026

Author: David J. Fox, ORCID 0009-0008-1290-6105

SCOPE: This directory contains Lean proof stubs for the P vs NP (BDP) Tower that were superseded by the certified computational approach in bdp1.out through bdp4.out. These stubs are HISTORICAL only and are not part of any Clay Millennium submission.

FILES ARCHIVED HERE:

(none -- all BDP Lean work migrated to src/M_FINAL/BDP_PhaseReversal.lean)

ACTIVE BDP LEAN FILE:

morningstar_spacecraft/src/M_FINAL/BDP_PhaseReversal.lean

-- Contains 23 'sorry' instances, all annotated as sorry-fills

-- pointing to certified .out files (bdp1.out through bdp4.out).

-- These sorries are in the P vs NP Tower ONLY.

-- They are NOT in the BSD submission scope (opera-numerorum/proofs/).

BSD SUBMISSION SCOPE:

opera-numerorum/proofs/ACT1-5/ -- SORRY: 0

opera-numerorum/proofs/ACT3/BSD_M22.lean -- M_star = 4/55, SORRY: 0

CLAY SEAL (frozen):

4. AXIOM INVENTORY: BSD uses only classical Lean axioms

4a. Custom 'axiom' declarations in proofs/ and src/CLAY/:

RESULT: PASS -- No custom axiom declarations found

4b. Classic trio (Lean 4 kernel built-ins, used implicitly):

propext : axiom propext : {a b : Prop} -> (a <=> b) -> a = b

Classical.choice : axiom Classical.choice : {a : Sort u} -> Nonempty a -> a

Quot.sound : axiom Quot.sound : {a : Sort u} -> {r : a -> a -> Prop}

-> {a b : a} -> r a b -> Quot.mk r a = Quot.mk r b

Source: Init.Prelude (Lean 4 kernel built-in, non-negotiable)

4c. Verification command (run after lake build):

#print axioms BSD_M22.M_star_eq_four_over_55

Expected: [propext, Classical.choice, Quot.sound]

5. CLAY SEAL: BSD computations frozen at M7 manifest

5a. CLAY seal construction:

Method: SHA256(cat m1.out m2.out m3.out m4.out m5.out m6.out)

Expected: 5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcbf7ebe3c9

Computed: 5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcbf7ebe3c9

RESULT: PASS -- CLAY seal matches frozen manifest

5b. Individual module SHAs (M1-M6):

m1.out: 63ef870a78766619327e99b68683bceff8c8ef9a525298756c77c8378fd2c291

m2.out: 3716c7dbb32524074b8fffb65eea45069c8b568a31dc73706405116b84029a83

m3.out: e687bb09a55e4eda198d4c5b24d03b7579f93bba27184a61fec7cbe29a83d044

m4.out: b810a7a331e47066e3eb4765a5ffdc17c1a56ddbff855a096c18ce2e9e2a19ed

m5.out: 9df98a3970acbb6942770a6cdd42fb21b009f9a5f45a222dd963e98ba4cb7a13

m6.out: ec9fa8c3aad478312c7e0d7373904dc3407eb5e9f4c19a011e3ca2ccb84da9fb

5c. CLAY_M1_M7.zip (archive of m1..m6.out):

Path: src/CLAY/CLAY_M1_M7.zip
SHA-256: 849482259416135553bfcec32bc0b60700a3eac3a9a38229ed803d46fc47f0a0
(Note: zip SHA differs from manifest SHA by design --
manifest SHA = SHA256(concatenated file contents),
zip SHA = SHA256(zip container format))

6. FORMAL DECLARATION

We certify that:

1. The BSD proof in `opera-numerorum/proofs/` imports no code from `morningstar_spacecraft` (`MORNING_STAR_REPO/`) or any spaceship tower.

2. The constant $M^* = 4/55$ is derived within `proofs/ACT3/BSD_M22.lean` via the `off_cliff_ratio` definition (`m22.out` SHA 5a5a345f...) and is not copied from any `morningstar_spacecraft` file.

3. The BSD submission scope (`proofs/`) is SORRY: 0.
`MORNING_STAR_REPO/` contains 23 annotated sorry-fills in `BDP_PhaseReversal.lean` (P vs NP Tower only). These are outside the BSD proof chain and do not affect BSD logical independence.

4. The BSD submission uses only the three classical axioms of Lean 4:
`[propext, Classical.choice, Quot.sound]`. No custom axioms.

5. CLAY computations are frozen at SHA256 =
5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fc7e7e3c9
covering M1 (`alpha_0`), M2 (`kappa`), M3 (`CF pi/10`),
M4 (`S14 primes`), M5 (`Bost sum`), M6 (`GRH X0(143)`).

Therefore, the BSD rank submission is logically independent of the `morningstar_spacecraft` repository, including RH Tower, MS Tower, and P vs NP Tower.

OVERALL STATUS: PASS

(Findings: `BDP_PhaseReversal.lean` sorries in `src/M_FINAL/` -- P vs NP Tower only, outside BSD proof scope. See Section 3 for details.)

Signed: David J. Fox, ORCID 0009-0008-1290-6105

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